PRODUCT DATASHEET · XR · XR-16

Brain & Face Signals in Virtual Reality

A brain-computer interface for VR headsets. A silicone face mask with 10 dry Ag/AgCl electrodes replaces the headset's facial interface, and a small BCI box on the head strap — built around the IronBCI-16 module — streams the signals over Bluetooth LE to the headset, a phone, a browser or a PC.



8 / 16	10	250 SPS	BLE 5	1.0 µVpp
Channels · XR / XR-16	Dry Ag/AgCl electrodes	Per channel	STM32WB55	Input noise

01 OVERVIEW

Key features

- **Fits the headset**: the mask clips in with four buckles in place of the facial interface, or use the silicone insert alone.
- **No gel needed**: 10 flat dry Ag/AgCl electrodes on the forehead and temples; a little gel or saline steadies the signal while moving.
- **IronBCI-16 inside**: two TI ADS1299 converters (24-bit) and an STM32WB55 Bluetooth controller on a Ø 50 mm two-board stack.
- **Wireless and battery-powered**: 5 V over USB-C from a power bank or a battery head strap; data over Bluetooth LE 5.
- **Open software**: BodyPress on Quest and Android, PiEEG Cloud in the browser, PiEEG Server with LSL, WebSocket and VRChat OSC.

Typical applications

- Neural-adaptive avatars (WebXR, VRChat)
- Attention and relaxation in VR
- Blink, gaze and expression control
- Face Trainer and VR games
- Research on EEG during movement
- Teaching with real biosignals

02 TECHNICAL SPECIFICATIONS

ACQUISITION	
Channels	8 (PiEEG XR) or 16 (PiEEG XR-16), plus reference and bias
BCI module	IronBCI-16: ADC board + MCU board, Ø 50 mm stack
Analog front end	2 × Texas Instruments ADS1299, 24-bit delta-sigma, simultaneous sampling; the 8-channel mask uses ADS1299 #1
Sample rate	250 SPS per channel
Noise	1.0 µVpp (input referred)
Signals	EEG (frontal), EOG (blinks, eye movement), EMG (facial expressions)
FACE MASK	
Electrodes	10 flat dry Ag/AgCl: 8 channels, reference, bias · 10 holes, Ø 1 mm
Positions	Forehead and temples (XR-PIN-004)
Material	Silicone; works dry, with gel or with saline
Fit	Made for one headset model; four buckles, or the silicone insert alone in the original interface
WIRELESS	

For research and education only. Not a medical device. Specifications subject to change.

Microcontroller	ST STM32WB55 (Cortex-M4 application core + Cortex-M0+ radio core)
Radio	Bluetooth LE 5, 2.4 GHz chip antenna
Advertised name	`PiEEG XR`
Protocol	IronBCI protocol: 24-bit big-endian samples on one notify characteristic (XR-SW-006)
POWER & SAFETY	
Supply	5 V over USB-C on the BCI box, from a power bank or a battery head strap
Isolation	Battery operation only while worn. Never a wall charger or a mains-powered computer
Classification	Research device. Not a medical device
COMPATIBILITY & SOFTWARE	
Headsets	Meta Quest 3, Meta Quest, PICO and others — each unit is made for one model
Software	BodyPress (Quest, Android) · PiEEG Cloud · PiEEG Server · VRChat OSC

03 SYSTEM DIAGRAM

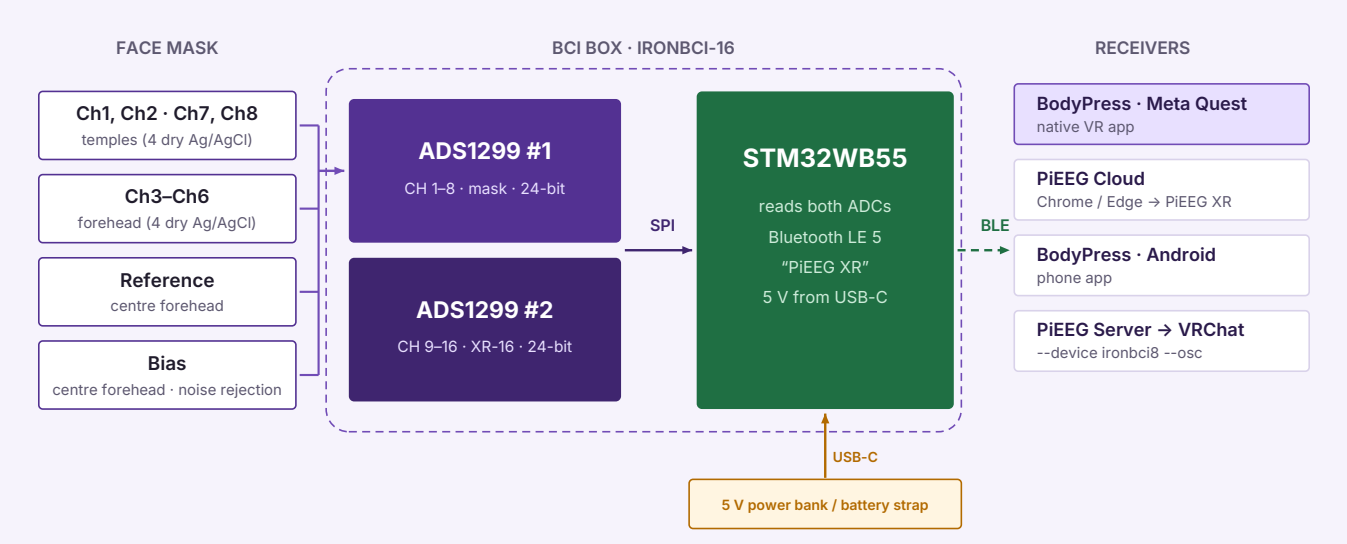


Figure 1 — Signal path. The mask electrodes connect by cable to the IronBCI-16 module in the BCI box, which streams the samples over Bluetooth LE.

04 EXAMPLE RECORDING

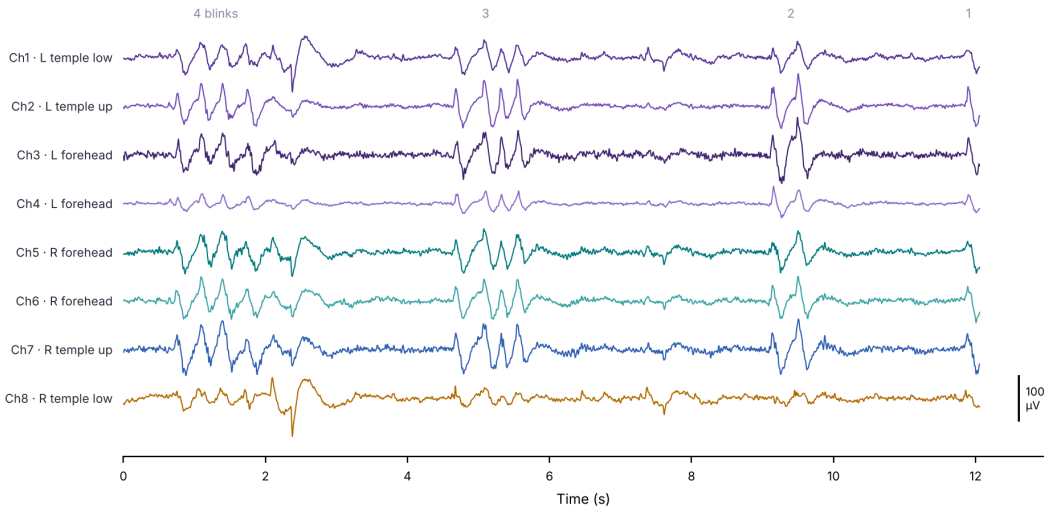


Figure 2 — Datasets/Data Band Pass (1-40 Hz) EEG Blinking 4-3-2-1.csv: blinking 4, 3, 2 and 1 times, dry electrodes, all 8 channels.

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05 AVAILABILITY

Product	PiEEG XR (8 channels) · PiEEG XR-16 (16 channels)
Order	pieeg.com · pieeg@pieeg.com — state your headset model
Repository	github.com/pieeg-club/PiEEG_XR_Hardware — firmware, datasets, VRChat manual
Citation	Y. El Abbassi, I. Rakhmatulin. PiEEG XR: A WebXR Brain-Computer Interface Platform for Neural-Adaptive Avatar Control in Mixed Reality. Preprints 2026, 2026060016.

Included	
Silicone face mask	10 dry electrodes and cable
BCI box	IronBCI-16 module, programmed
ST-Link V2	With jumper wires, for firmware updates

Required, not included	
VR headset	The model the unit is made for
5 V source	Power bank or battery head strap
USB-A to USB-C cable	Most reliable for power
Receiver	Headset, Android, Chrome / Edge or a PC

WARNING Battery only.

Power the BCI box from a 5 V power bank or a battery head strap — never from a wall charger or a computer on mains power, and do not charge the headset while wearing PiEEG XR. PiEEG XR is not a medical device.

06 RELATED DOCUMENTS

XR-MD-002 Mechanical & Assembly Drawing	**XR-PCB-003** PCB & Manufacturing Specification	**XR-PIN-004** Mask, Electrode & Connector Guide
XR-QSG-005 Quick Start & Safety Guide	**XR-SW-006** Bluetooth Interface & Software Guide	